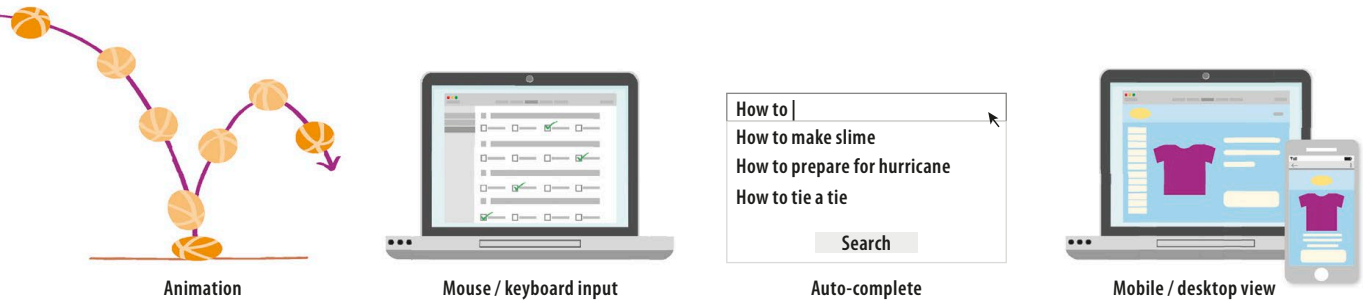


Growth of JavaScript

Professional developers looked down on JavaScript at first, largely because it was designed for and mainly used by amateur programmers. Also, unlike Java applications, JavaScript programs couldn't move data to and from a database on a server. This changed with the introduction of AJAX (Asynchronous JavaScript And XML), a collection of web technologies, including JavaScript, that allowed webpages to connect to a server. Professionals now consider JavaScript to be a useful language, and lots of code libraries have been written to make a variety of tasks easier.

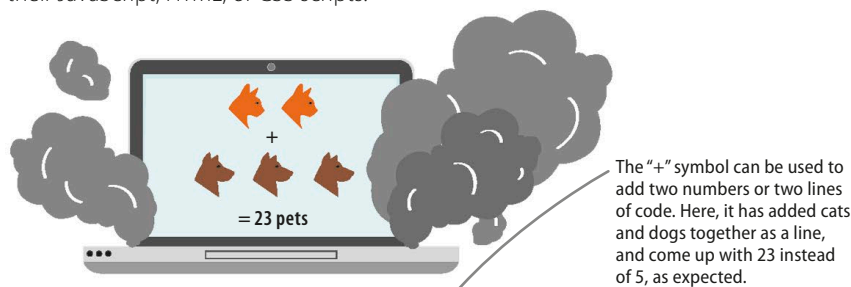
▽ Using the console

JavaScript programs are extremely versatile. They allow programmers to create code that will help with multiple aspects of websites, such as animation, user input, auto-complete technology, and enabling the smooth flow of user interface from desktop to mobile websites.



Why JavaScript?

For programmers new to web development, JavaScript is a great place to start. Using just a text editor and a web browser, programmers can add an array of interactivity and animations to their sites, from the very simple to the quite complex. It's also possible to create apps, known as web apps, using only JavaScript, HTML, and CSS. These apps run on any mobile phone with a browser. There are also websites that help new programmers by allowing them to see the real-time effects of changing their JavaScript, HTML, or CSS scripts.



```
function pets() {
  var cats = 2;
  var dogs = "3";
  console.log("Number of pets:" + (cats + dogs));
}
```

△ Drawbacks

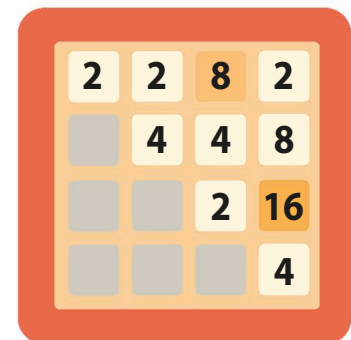
A variable's type describes whether it's a number, character, or something else. JavaScript doesn't have strict rules about types, which can sometimes lead to unexpected results.

This code stores numerical values for cats and dogs and then asks to add them together.

REAL WORLD

JavaScript games

Many browser-based games, including the original version of 2048 – a popular number-puzzle game – are written in JavaScript. Several free JavaScript game engines are also available, which allow programmers to easily create browser-based games. There are even games where the player has to write some JavaScript to complete each level.



A game of 2048